

Bridge Circuits

DC bridges

1. Wheatstone bridge

- **The Wheatstone bridge** consists of **only resistances** in the respective branches; it may be used with either ac or dc excitation.
- **The Wheatstone bridge** is used mainly for **resistance measurements**. for measuring unknown resistance by balancing two branches of a bridge circuit.

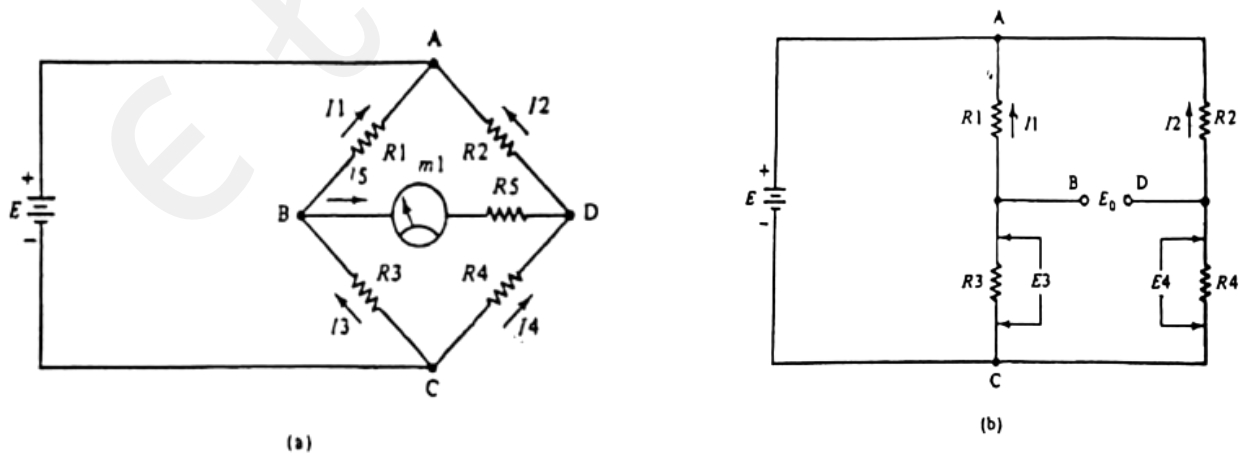


Fig. (2-1) (a) The Wheatstone bridge, (b) redrawn to make more clear.

Current I_5 through the galvanometer is given by

$$I_5 = \frac{E_0}{R_5 + R_m}$$

where I_5 = the current flowing in meter M in amperes (A)

E_0 = the potential in volts (V) between points B and D

R_5 = the resistance of resistor R_5 in ohms (Ω)

R_m = the resistance of the meter movement in ohms (Ω).

Voltage E_0 is the difference between voltages E_3 and E_4 :

$$E_0 = E_3 - E_4 \quad (2-2)$$

But E_3 and E_4 are given in terms of E and the resistances:

$$E_3 = \frac{ER_3}{R_1 + R_3} \quad (2-3)$$

And

$$E_4 = \frac{ER_4}{R_2 + R_4} \quad (2-4)$$

By substituting Equations (1-3) and (1-4) into (1-2), we obtain

$$E_0 = E \times \left[\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right] \quad (2-5)$$

Assuming that R_m is negligible, and substituting Equation (2-5) into (2-1), we have

$$I_5 = \left[\frac{E}{R_5} \right] \left[\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right] \quad (2-7)$$

- **The null condition** For a **DC Wheatstone bridge** when the bridge is perfectly balanced, meaning that there is no potential difference between the midpoints of the two branches where the galvanometer is connected. As a result, no current flows through the galvanometer.

$$\frac{R_1}{R_3} = \frac{R_2}{R_4}$$

$$R_x = \frac{R_2 R_3}{R_1}$$

(when the bridge is in null).

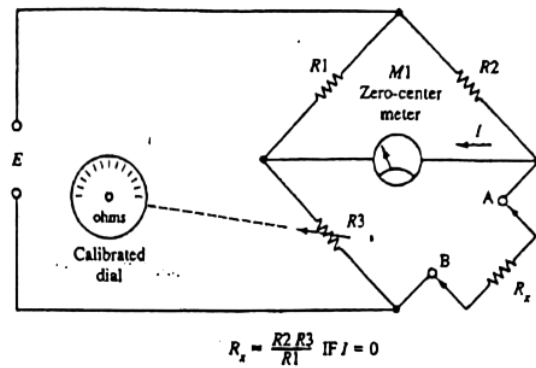


Fig. (2-22) Measuring resistance with a Wheatstone bridge.

- **The zero-center galvanometer** is used as a **null detector** in **dc Wheatstone bridges**.

ac bridges

Alternating-current current meters or ac voltmeters are used as **null detectors** in ac bridges when there are **no phase components** in the output signal, while an **oscilloscope or a phase detector** can be used when there are both **phase and magnitude components**.

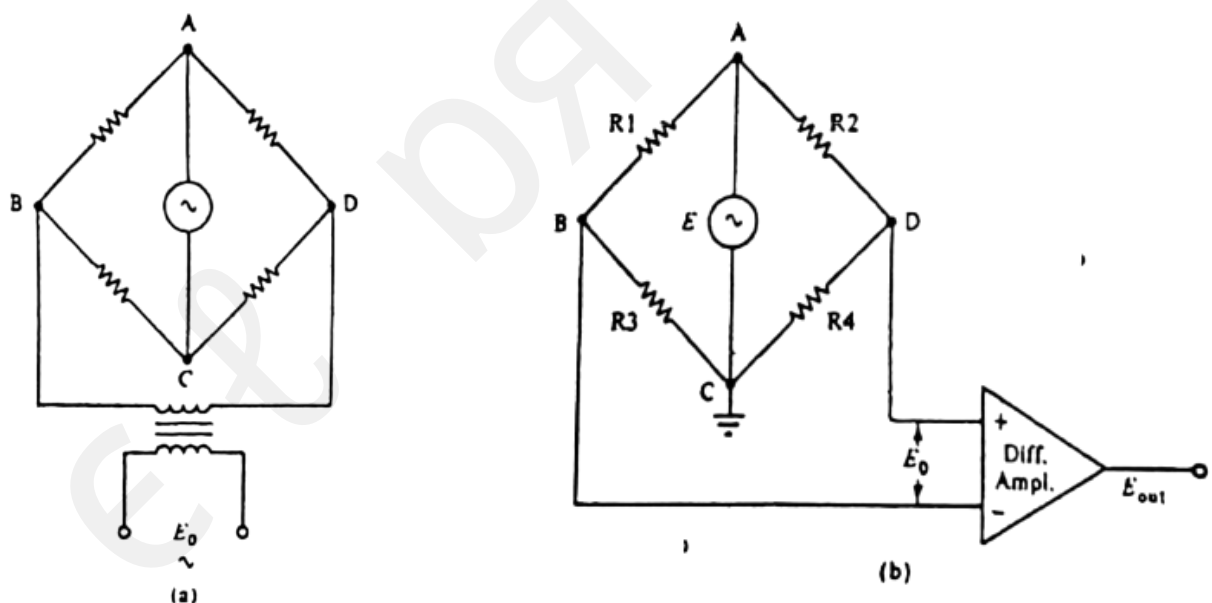


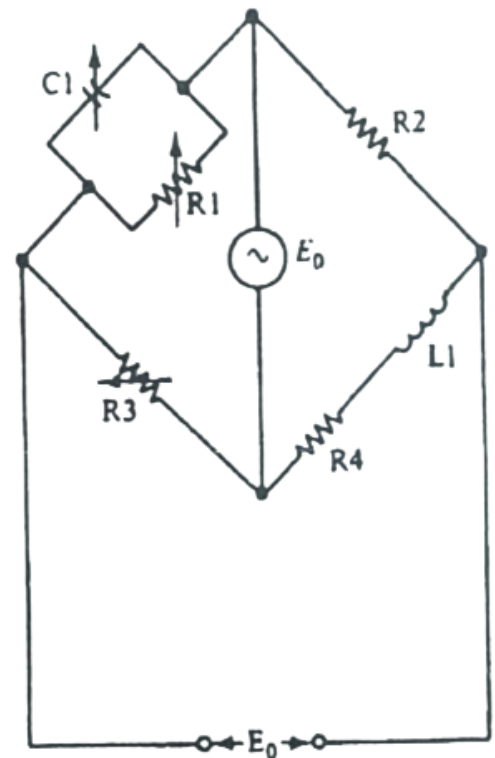
Fig. (2-5) (a) Transformer-coupled ac Wheatstone bridge, (b) differential amplifier output for an ac Wheatstone bridge.

Types of ac bridges

1. Maxwell's bridge

The null condition for a Maxwell bridge circuit is given by the following equations:

$$\begin{aligned} - L1 &= R2 \times R3 \times C1 \\ - R4 &= \frac{R2 \times R3}{R1} \end{aligned}$$



- The Maxwell bridge measures **unknown inductance values** (e.g., L1) with balance equations independent of frequency.
- It is **not sensitive to resistive losses** and easier to calibrate capacitors for C1 than inductors for L1.
- The principal use of the Maxwell bridge is **inductance measurement**, often in Q-meters for quality factor Q of inductors, though Q is frequency sensitive.

$$L1 = R2 R3 C1$$

$$R4 = \frac{R3}{R1} \times R2$$

$$Q_L = \omega C1 R1$$

$$Q_L = \omega \times R1 \times C1$$

where $\omega = 2\pi f$

R_1 and C_1 are as previously defined.

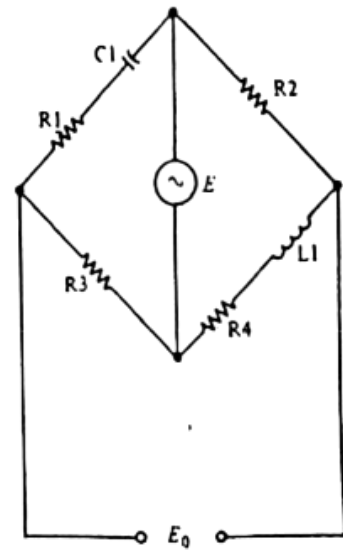
2. The Hay bridge

The Hay bridge circuit is like Maxwell's bridge, but with the R_1/C_1 combination connected in series.

The Hay bridge **is frequency sensitive**.

The balance equations are as follows:

$$\begin{aligned} - L1 &= \frac{R2 \times R3 \times C1}{1 + \left[\frac{1}{Q}\right]^2} \\ - R4 &= \left[\frac{R2 \times R3}{R1}\right] \times \left[\frac{1}{Q^2 + 1}\right] \\ - Q &= \frac{1}{\omega \times R1 \times C1} \end{aligned}$$



$$R4 = \frac{R2 R3}{R1} \times \frac{1}{Q^2 + 1} \quad L1 = \frac{R2 R3 C1}{1 + \left(\frac{1}{Q}\right)^2}$$

$$Q = \frac{1}{\omega R1 C1}$$

- **The Hay bridge** is frequency-sensitive and **preferred for high-Q inductors**, while **the Maxwell bridge** works best with **low-Q inductors**.
- **A frequency-independent version of The Hay bridge** can be used when Q is very high (greater than about 10), simplified as:

$$L1 = R2 \times R3 \times C1$$

Using this **approximation** results in **1% error if Q is 10** and **0.01% if Q is 100**.

Proof Let .

$$R2 C3 C1 = k = \text{constant}$$

Assume

$$Q = 10$$

a. From Equation (2-20):

$$\begin{aligned} L1 &= k[1 + (1/Q)^2] \\ &= k[1 + (1/10)^2] \\ &= k(1 + 0.01) = k/1.01 = 0.99k \end{aligned}$$

b. From Equation (2-22):

$$L1 = k$$

$$c. \frac{0.99k - k}{0.99k} \times 100 = -1\%$$

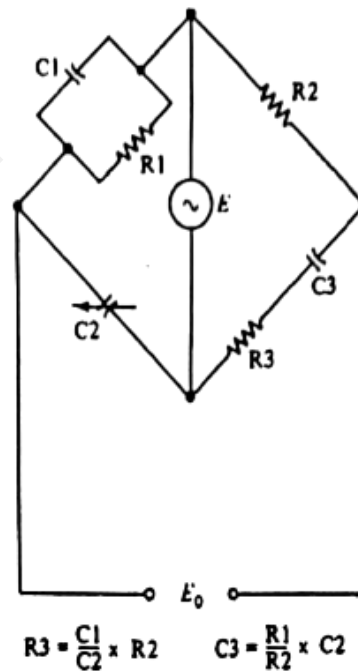
3. The Schering Bridge

As shown in next Fig.

The balance equations are:

$$- R_3 = \frac{R_2 \times C_1}{C_2}$$

$$- C_3 = \frac{C_2 \times R_1}{R_2}$$



The Schering bridge is used primarily for measuring capacitance and the power factor of capacitors, the Schering Bridge's Q for the capacitor is found from:

$$Q_{C3} = \frac{1}{\omega \times R_1 \times C_1}$$

Summary :

1. **The Wheatstone bridge** consists of **only resistances** in the respective branches; it may be used with either ac or dc excitation.
2. **The Maxwell bridge** and **the Hay bridge** are used for **inductance measurements**, whereas **the Schering bridge** is used for **capacitance measurement**. **The Wheatstone bridge** is used mainly for **resistance measurements**.
3. **The zero-center galvanometer** is used as a null detector in dc Wheatstone bridges.
4. **Alternating-current current meters or ac voltmeters** are used as **null detectors** in ac bridges when there are **no phase components** in the output signal, while an **oscilloscope or a phase detector** can be used when there are both **phase and magnitude components**.